



Data Mining

Final Review

<https://data-mining.github.io/winter-2026/>

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Neural Network Architecture

1. Dimension: input/output based on concrete application
2. Physical Meaning of Neural Network: WX , describe and draw the picture justifying the physical meaning
3. Why we need to have WX , $WX+B$, nonlinearity function
4. **Linear Regression**
 1. **Problem Formulation**
 2. **Understand Loss Function and how optimize the loss will lead to the desired effect we want**
 3. **Theoretical/Computational Optimization**
 4. **Local/Global Optimal?**
5. **Classification**
 1. **Binary Classification Loss, understand why optimizing the loss lead to the desired effect we want**
 2. **Multi-Class Classification Loss, understand why optimizing the loss lead to the desired effect we want**





Graph Mining

1. How to represent a graph-structured data typically in ML?
 1. Feature Matrix
 2. Graph Adjacency Matrix, Graph Adjacency List, Pros/Cons?
2. **Mathematically represent message-passing**
 1. **Understand the physical meaning**
 2. **Can compute concrete example**
3. Differences between Label Propagation, MLP and GNN in node classification
 1. Key assumption in label propagation and GNN
 2. Difference between label propagation and GNN
4. Mathematical Format of GNN
5. Differences between Heuristic Method, MLP, and GNN in link prediction
 1. Common Neighbor/Jaccard Similarity



Image Mining

1. How to represent an Image-structured data in computer?
2. **Problem of MLP for Image Classification**
 1. **Able to work for Image Translation? If not, why?**
 2. **Able to work for Image Rotation? If not, why?**
3. **Invariance and Equivariance, can you define and identify whether a specific operator is Invariance or Equivariance?**
4. CNN Architecture
 1. Kernal Convolution Operation
 1. Stride
 2. Padding
 3. Dimension Computation after applying Kernal Convolution
 2. Pooling (Max/Mean/...)
5. Explanation
 1. Principal of Grad-CAM
 2. Principal of Using Mask to Explain



Language Mining

1. Language Mining Application and how to represent an Language-structured data in computer?
- 2. General Formulation of Next Token Generation**
 - 1. Notation**
 - 2. Conditional Distribution**
 3. Distribution Factorization
3. N-Gram Language Model
 1. Core Technique (Frequency Computing)
 2. Pros/Cons
4. RNN
 - 1. Token Embedding, Word2Vec**
 - 2. How to query token embedding given token ID?**
 - 3. Architecture of RNN and Physical Meaning of Each Component**
 4. How we want to aggregate the output to do downstream task
 1. Last one?
 2. Every layer output?
 5. Problem with RNN: Gradient Vanish and Gradient Explosion





Transformer and LLM

1. Motivation of Transformation

2. Attention:

1. Motivation of Attention

2. Physical Meaning of Query, Key and Value

3. Master Every Detail and I will let you physical compute the forward process of Attention

4. Multi-Head

5. Causal Attention: Which Task?

6. Cross Attention: Which Task?

7. Self Attention: Which Task?

3. Positional Encoding

4. What is auto-regressive generation?

5. What is beam search?

6. How to train LLM, what is the loss? Physical Meaning

